**Solution:**

Distance travelled by Madhan from

Coimbatore to Karur via Vellakoil = 131 $km$

Distance travelled by Madhan from

Karur to Coimbatore via Erode = 66 + 101 = 167 $km$

Therefore, the route via Vellakoil is the shortest route.

## Example 6.4

A telephone exchange worker wants to give the telephone connection to Amutha's house. Fig.6.27 shows all the possibilities of cable connections. Find the route which requires minimum cable?

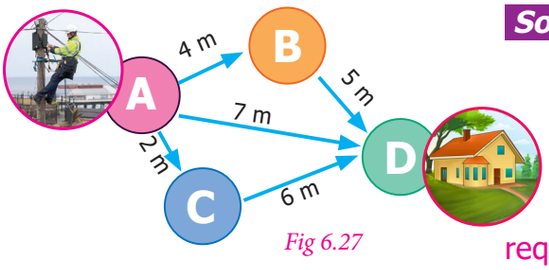



*Fig 6.27*

***Solution***

Distance from A to D =7 $m$

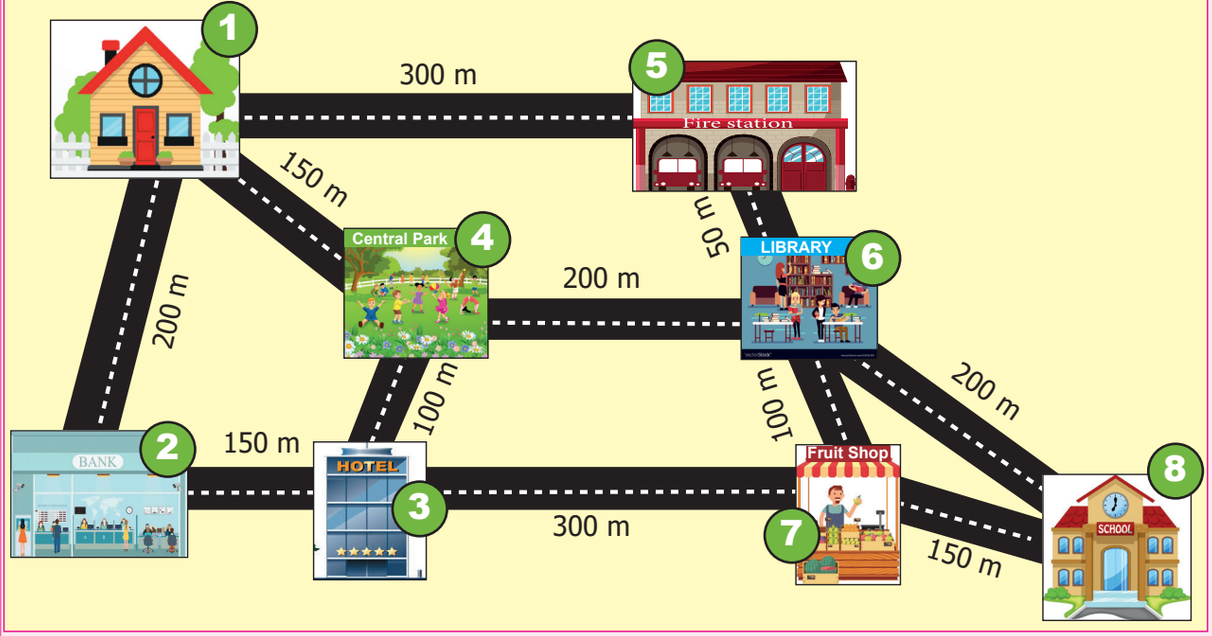
Distance from A to D (via B) = 4+5 = 9 $m$

Distance from A to D (via C) = 2+6 = 8 $m$

Therefore, the route directly from A to D requires minimum cable of length 7 $m$.

**Try these**

1. Observe the picture and answer the following




(i) Find all the possible routes from house to school via fire station

(ii) Find all the possible routes between central park and school with distance. Mention the shortest route?

(iii) Calculate the shortest distance between bank and school.



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